

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

x	$g(x)$
1	54
2	51
3	48
4	45

For the linear function g , the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b ?

Answer

- A. 3
- B. 27
- C. 54
- D. 57

Correct Answer: D

Rationale

Choice D is correct. It's given that for the linear function g , the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 54. Substituting 1 for x and 54 for $g(x)$ in $g(x) = mx + b$ yields $54 = m(1) + b$ or, $54 = m + b$. Subtracting b from both sides of this equation yields $54 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 51. Substituting 2 for x and 51 for $g(x)$ in $g(x) = mx + b$ yields $51 = m(2) + b$, or $51 = 2m + b$. Substituting $54 - b$ for m in this equation yields $51 = 2(54 - b) + b$. Applying the distributive property to the right-hand side of this equation yields $51 = 108 - 2b + b$, or $51 = 108 - b$. Subtracting 108 from both sides of this equation yields $-57 = -b$. Dividing both sides of this equation by -1 yields $57 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.